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Part I

Theoretical Background

Chapter 1

Mori-Zwanzig theory

The Mori-Zwanzig theory allows to observe the evolution of a system through the projection on a subset of its degrees of freedom, the evolution of the selected variables is determined by the sum of a deterministic part and a random, "noisy" part. The randomness reflects one's ignorance of unresolved degrees of freedom. This is done via a projection operator technique getting the *generalized Langevin equation* (GLE), an equation of the form

$$\frac{dA}{dt} = i\Omega A(t) - \int_0^t d\tau K(\tau)A(t-\tau) + F(t) \tag{1.1}$$

where Ω, $K(t)$ and $F(t)$ are well-defined functions, called the frequency, memory function, and random force respectively. This equation describes the non-equilibrium evolution of any set of functions $A(t)$ defined on the phase-space of the microscopic system and, depending on the context, these functions have been referred to as collective variables, relevant variables, coarse-grained variables or collective modes.

This theory has been know in the context of transport theory and for example in the dynamic light scattering, the GLE is used to compute time-correlation functions under circumstances where the usual phenomenological equations do not apply. Indeed throught the equation (1.1) can be obtained the equation for the time correlation function $C(t)$:

$$\frac{dC(t)}{dt} = i\Omega C(t) - \int_0^t d\tau K(\tau)C(t-\tau)\,. \tag{1.2}$$

This is called the *memory-function equation* and the random force appears implicitly in $K(t)$ [1].

The projection operator technique are powerful tools for the development of a statistical-mechanical foundation of irreversible processes in many-body systems [3]. These tools has been introduced into statistical physics by S. Nakajima in 1958 and R. Zwanzig in 1960. The former to study the quantum statistical mechanics of steady diffusion [7], and the latter to give a new formulation of methods introduced in the statistical-mechanical theory of irreversibility [11]. Later in 1965, H. Mori proposed an projection operator technique in Heisenberg picture, which leads to the linear GLE for arbitrary systems close to thermal equilibrium [6]. Some years later in 1973 R. Zwanzig derived a class of generalized Langevin equations describing the motion of a system interacting with a

heat bath, this GLE is exact but differs from that obtained by Mori's scheme for two reasons: for the presence of an arbitrary nonlinear force and for the fact that its validity is not restricted to small deviations from the thermal equilibrium [12]. In recent years, D. Kauzlarić et all showed that Mori GLE can be obtained from the Zwanzig's one if the system is close to the equilibrium and under the Markovian assumption, i.e. that the relevant variables, the functions $A(t)$ in the equation (1.1), evolve in a time scale much larger than the memory time scale [4].

Chapter 2

Numerical Method

2.1 Langevin Dynamics Integrator: BAOAB method

In this work we simulated the Langevin Dynamics by the BAOAB method, which was derived and analysed by B. Leimkuhler and C. Matthews [5].

The aim of their work is to compare Langevin dynamics integrators in term of their invariant distributions and demonstrate a superconvergence property of a certain method in the high friction limit (or in the null mass limit). Numerically they implemented the various methods considered and compared the accuracy of configurational sampling for different values of the friction parameter γ and a range of timesteps.

Their results show that the best method to sample the configurational Gibbs-Boltzmann distribution[1] is the BAOAB method and in the context of molecular dynamics it is a cheap and easy to implement scheme using only a single force vector per timestep:

$$
\begin{aligned}
p_{n+1/2} &= p_n - \frac{1}{2}\nabla U\left(x_n\right)\, dt\,; \\
x_{n+1/2} &= x_n + \frac{1}{2}\, M^{-1}\, p_{n+1/2}\, dt\,; \\
\hat{p}_{n+1/2} &= c_1\, p_{n+1/2} + c_2\, M^{1/2}\, R_{n+1}\,; \\
x_{n+1} &= x_{n+1/2} + \frac{1}{2}\, M^{-1}\, \hat{p}_{n+1/2}\, dt\,; \\
p_{n+1} &= \hat{p}_{n+1/2} - \frac{1}{2}\nabla U\left(x_{n+1}\right)\, dt\,.
\end{aligned}
$$

Here, R_i is an N-vector of iid normal random numbers with mean 0 and variance 1, M is the diagonal mass matrix, dt is the timestep,

$$
c_1 = e^{-\tilde{\gamma}\, dt} \quad \text{and} \quad c_2 = \sqrt{\left(1 - c_1^2\right)/\beta}
$$

are evaluated for a given friction parameter $\tilde{\gamma} = \gamma\, M^{-1}$ and a given temperature parameter $\beta = (K_B T)^{-1}$, where K_B is the Boltzmann's constant.

[1] This means the calculation of functions averages of phase space variables of a molecular N-body system modelled at constant temperature.

2.1.1 An idea of how the BAOAB method was derived

Background

An Itô stochastic differential equation (SDE) can be written as

$$d\mathbf{x}(t) = \mathbf{a}\left[\mathbf{x}(t), t\right] dt + \mathbf{b}\left[\mathbf{x}(t), t\right] d\mathbf{W}(t)\,,$$

where $\mathbf{x}$ are the n stochastic process, $\mathbf{a}(\mathbf{x}, t)$ and $\mathbf{b}(\mathbf{x}, t)$ are certain know functions, $d\mathbf{W} \equiv \boldsymbol{\xi}(t)\, dt$ is an n variable Wiener process and $\boldsymbol{\xi}(t)$ is the white noise. This SDE corresponds an Fokker-Planck equation if the stochastic process described by a conditional probability density $p(\mathbf{x}, t \,|\, \mathbf{x_0}, t_0) \equiv p$ satisfy

$$\partial_t\, p = \mathscr{L}(\mathbf{x}, t)\, p = -\sum_i \partial_i \left[A_i(\mathbf{x}, t)\, p\right] + \frac{1}{2}\sum_{ij} \partial_i \partial_j \left[B_{ij}(\mathbf{x}, t)\, p\right]$$

where $\mathscr{L}(\mathbf{x}, t)$ is a second-order differential operator and the coefficients of RHS are such that $\mathbf{A}(\mathbf{x}, t) = \mathbf{a}(\mathbf{x}, t)$ and $\underline{\mathbf{B}}(\mathbf{x}, t) = \mathbf{b}(\mathbf{x}, t)\mathbf{b}^T(\mathbf{x}, t)$. The Fokker-Planck equation describes diffusion processes, the vector $\mathbf{A}(\mathbf{x}, t)$ is know as drift vector and the matrix $\underline{\mathbf{B}}(\mathbf{x}, t)$ is know as the diffusion matrix. From its definition this matrix is positive semidefinite and symmetric [2].

Let us consider an Orstein-Uhlenbeck (OU) SDE, that is, an Itô equation where the coefficients are

$$\mathbf{a}(\mathbf{x}) = -k\,\mathbf{x} \quad \text{and} \quad b = \sqrt{D}$$

here k and D are positive parameters. The solution of one dimensional OU-SDE can be computed by the Itô's formula applied to $y = xe^{k\,t}$[2]. We get

$$x(t) = x_0(t_0)\, e^{-k(t-t_0)} + \sqrt{D}\int_{t_0}^{t} dW(t') e^{-k(t-t')}\,,$$

and the average and variance are respectively:

$$\mathbb{E}[x(t)] = \mathbb{E}[x_0(t_0)]\, e^{-k(t-t_0)}\,,$$
$$\mathbb{E}\left[\left(x(t) - \mathbb{E}[x(t)]\right)^2\right] = \left[Var(x_0) - \frac{D}{2k}\right] e^{-2k(t-t_0)} + \frac{D}{2k}\,.$$

We can easily show in the stationary limit ($t \gg t_0$) and applying the equipartition theorem that the diffusive parameter D is equal to $(2k)/\beta$.

Langevin Dynamics

Langevin dynamics combines a conservative dynamic with the OU stochastic vector field as a forcing for the momentum, so we are assuming that the stochastic force is not affected by the potential:

$$M\frac{d^2\mathbf{x}}{dt^2} = -\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\,\frac{d\mathbf{x}}{dt} + \sqrt{(2\gamma)/\beta}\,\boldsymbol{\xi}(t)$$

where we introduced physical quantities: M is the mass of the particle, $\mathbf{x}$ is the coordinate of position, γ is the friction parameter and $\beta = (K_B T)^{-1}$ is the temperature parameter. We can rewrite the equation as

$$\begin{cases} d\mathbf{x} = \mathbf{v}\, dt \\ M d\mathbf{v} = \left[-\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\,\mathbf{v}\right]\, dt + \sqrt{(2\gamma)/\beta}\, d\mathbf{W} \end{cases}$$

and for the canonical coordinates $\{\mathbf{q}(t), \mathbf{p}(t)\}$ we get

$$\begin{cases} d\mathbf{q} = M^{-1}\,\mathbf{p}\,dt \\ d\mathbf{p} = [-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p}]\;dt + \sqrt{(2\tilde{\gamma})/\beta}\,M^{1/2} d\mathbf{W} \end{cases}$$

where $\tilde{\gamma} = \gamma/M$. From the system of two first-order equations can we compute the drift and diffusion coefficients

$$\begin{aligned} (A_{\mathbf{q}})_i &= M^{-1}\,p_i\,; & (B_{\mathbf{qq}})_{ij} &= (B_{\mathbf{pq}})_{ij} = 0\,; \\ (A_{\mathbf{p}})_i &= (-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p})_i\;; & (B_{\mathbf{pp}})_{ij} &= \frac{2\tilde{\gamma}}{\beta}\,M\,\delta_{ij}\,; \end{aligned}$$

and thus determine the Fokker-Planck operator for the Langevin dynamics

$$\begin{aligned} \mathscr{L}_{LD}\,\varphi = &-M^{-1}\sum_i p_i\,\partial_{q_i}\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\sum_i \partial_{p_i}\,\varphi \;+ \\ &+\;\tilde{\gamma}\sum_i \partial_{p_i}\,p_i\,\varphi \;+\; \tilde{\gamma}\sum_i p_i\,\partial_{p_i}\,\varphi \;+\; \tilde{\gamma}\,K_B T\sum_i M\,\partial_i^2\,\varphi = \\ &= -M^{-1}\mathbf{p}\cdot\boldsymbol{\nabla}_q\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\cdot\boldsymbol{\nabla}_p\,\varphi \;+\; \tilde{\gamma}\,[\,\boldsymbol{\nabla}_p(\mathbf{p}\,\varphi) + K_B T\boldsymbol{\Delta}_p\,\varphi\,]\;, \end{aligned}$$

where $\boldsymbol{\Delta}_p$ is the mass-weighted Laplacian in the momenta and φ is a test function [9].

Splitting methods

2.1.2 Some more details about the work of B. Leimkuhler and C. Matthews

The methods that are considered are splitting integrators that decompose the stochastic vector field which drives the Langevin Dynamics into simpler vector fields which can be solved exactly. Nevertheless, the composition method that results cannot be directly related to a stochastic differential equation, but nonetheless the invariant measure associated to the numerical method can be derived, using the Baker-Campbell-Hausdorff (BCH) expansion, as an asymptotic series in two parameters: the stepsize and the coefficient γ^{-1}. The authors stress that their approach focuses on the truncation error of the invariant distribution, thus the greatest benefit would be seen only when the statistical error is well controlled. Furthermore, they observe that the least biased scheme from the point of view of the error introduced in configurational sampling is also as efficient as the alternatives and the most robust with respect the variation of parameters (stepsize and γ^{-1}), thus they emphasise that there is no price for the improvement in accuracy.

Appendix A

Elements of Probability Theory and Stochastic Processes

We introduce some basic definition and results from probability theory and stochastic processes. This material can be found in [8, 10].

Definition A.1 (**Sample Space**)**.** The set of all events, such as all the possible outcomes of an experiment, is called sample space and is denoted by Ω.

Definition A.2 (σ**-algebra**)**.** A collection $\mathscr{F}$ of subset of Ω is called σ-algebra on Ω if[1]

(i) $\Omega \in \mathscr{F}$;
(ii) $F \in \mathscr{F} \implies F^c := \Omega \setminus F \in \mathscr{F}$;
(iii) $F_n \in \mathscr{F}$, $n \in \mathbb{N} \implies \bigcup_{n\in\mathbb{N}} F_n \in \mathscr{F}$.

Intuitively, the σ-algebra $\mathscr{F}$ contains all the useful information that is available about the random experiment that we are performing. One of its elements is called a $\mathscr{F}$-measurable subset of Ω or *event*.

Definition A.3 (**Measurable Space**)**.** A pair $(\Omega, \mathscr{F})$, where Ω is a set and $\mathscr{F}$ is a σ-algebra on Ω, is called measurable space.

Definition A.4 (**Probability Measure**)**.** A probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$ is a function

$$\mathbb{P} : \mathscr{F} \to [0, 1];\ F \mapsto \mathbb{P}(F);$$

satisfying

(i) $\mathbb{P}(\Omega) = 1$;

[1] We use the following convention:

$$\mathbb{N} := \{1, 2, 3, \dots\} \subseteq \{0, 1, 2, 3, \dots\} =: \mathbb{Z}^+ \quad \text{and} \quad \mathbb{R}^+ := [0, +\infty)\,.$$

(ii) if $F_n \in \mathscr{F}$ with $F_n \bigcap F_m = \emptyset$, $n \neq m$ and $n, m \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} F_n\right) = \sum_{n\in\mathbb{N}} \mathbb{P}\left(F_n\right) .$$

.

Definition A.5 (**Probability Space**)**.** A probability space is a triple $(\Omega, \mathscr{F}, \mathbb{P})$ comprising a set Ω, a σ-algebra $\mathscr{F}$ of subsets of Ω and a probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$.

Since we are usually interested in the consequence of the outcomes of an experiment, we introduce the notion of random variable that is intuitively a function of the outcomes. We do it following the Williams [10], through the definition of measurable function from $(\Omega, \mathscr{F}, \mathbb{P})$ to $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$[2].

Definition A.6 (**$\mathscr{F}$-measurable function**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ and $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be two measurable spaces. The function $\psi : \Omega \to \mathbb{R}$ is called $\mathscr{F}$-measurable function if

$$\psi^{-1}(A) \coloneqq \{\omega \in \Omega : \psi(\omega) \in A\} \in \mathscr{F}, \quad \forall\, A \in \mathscr{B}(\mathbb{R}) .$$

Definition A.7 (**Random Variable**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A random variable is $\mathscr{F}$-measurable function, thus

$$X : \Omega \to \mathbb{R} , \quad X^{-1} : \mathscr{B}(\mathbb{R}) \to \mathscr{F} .$$

Every random variable, carried by some probability triple $(\Omega, \mathscr{F}, \mathbb{P})$, induces a probability measure $\mathbb{P}_X$ on $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$:

$$\begin{aligned} \mathbb{P}_X : \mathscr{B}(\mathbb{R}) &\to [0,1] \\ B &\mapsto \mathbb{P}_X(B) \coloneqq \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)) . \end{aligned}$$

The measure $\mathbb{P}_X$ is called *distribution* (or sometimes the *law*) of X. We can now define the *distribution function* of X:

$$\begin{aligned} F_X : \mathbb{R} &\to [0,1] \\ x &\mapsto F_X(x) \coloneqq \mathbb{P}\left(\{\omega \in \Omega : X(\omega) \leq x\}\right) = \mathbb{P}\left(X \leq x\right) . \end{aligned}$$

This function F_X indicates the likelihood that the random variable X is in $(-\infty, x]$.

Definition A.8 (**Probability density function**)**.** Let X a random variable and $f : \mathbb{R} \to \mathbb{R}$ a function such that

$$f(x) \geq 0 \quad \forall\, x \in \mathbb{R} \qquad \text{and} \qquad \int_{\mathbb{R}} f(x)\, dx = 1 .$$

We say that X has a *probability density function* (pdf) f_X if

$$\mathbb{P}(X \in B) = \int_B f_X(x)\, dx \qquad \forall\, B \in \mathscr{B}(\mathbb{R})$$

[2]We will use the notation $\mathscr{B}(\mathbb{R})$ to denote the Borel σ-algebra of $\mathbb{R}$, the smallest σ-algebra containing all open sets of $\mathbb{R}$.

Definition A.9 (**Expected value**)**.** We define the *expectation* $\mathbb{E}[X]$ of a random variable X by

$$\mathbb{E}[X] \coloneqq \int_\Omega X \, d\mathbb{P} \equiv \int_\Omega X(\omega)\, \mathbb{P}(d\omega) = \int_{\mathbb{R}} x \, \mathbb{P}_X(dx)\,.$$

In the last equality we changed variable $x = X(\omega)$ and used the definition of distribution of X, that is $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$.

Let $h : \mathbb{R} \to \mathbb{R}$ a Borel measurable function. Then

$$h(X) \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P}) \coloneqq \left\{ X \in (\Omega, \mathscr{F}, \mathbb{P}) : \ \mathbb{E}\left[|X|^1\right] < \infty \right\} \ \Leftrightarrow \ h \in \mathcal{L}^1(\Omega, \mathscr{B}, \mathbb{P}_X)$$

and then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\, \mathbb{P}_X(dx)\,.$$

Furthermore, if f_X is a pdf of X and

$$\mathbb{E}[|h(X)|] < \infty \iff \int_{\mathbb{R}} |h(x)|\, f_X(x)\, dx < \infty\,,$$

then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\, f_X(x)\, dx\,.$$

Appendix B

Liouville Space

In classical mechanics, the phase-space is the space of all possible states of a physical system composed by M particles. The state is represented by the point $\boldsymbol{\gamma}$ specified by the vector

$$\boldsymbol{\gamma} = \left(\boldsymbol{\gamma}_{\mathbf{r}}, \boldsymbol{\gamma}_{\mathbf{p}}\right) = \left(\mathbf{r}_1, \ldots, \mathbf{r}_M, \mathbf{p}_1, \ldots, \mathbf{p}_M\right) .$$

The evolution of a state is obtained in according to the canonical equations of motion

$$\begin{aligned} \frac{\mathrm{d}\boldsymbol{\gamma}}{\mathrm{d}t} &= \sum_{i=1}^{M} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \dot{\mathbf{r}}_i + \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \dot{\mathbf{p}}_i \right] = \\ &= \sum_{i=1}^{M} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} - \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \right] \equiv \{\boldsymbol{\gamma}, \mathcal{H}\} \end{aligned} \tag{B.1}$$

where $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is Poisson bracket of $\boldsymbol{\gamma}$ with the Hamiltonian $\mathcal{H}$. The equation obtained indicates that $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is a generator of the time evolution of $\boldsymbol{\gamma}(t)$.

The Poisson bracket allows us to introduce an operator L called *Liouvillian* or *Liouville operator*

$$iL \equiv \{\ldots, \mathcal{H}\} = \sum_{i=1}^{M} \left[\frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \cdot \frac{\partial}{\partial \mathbf{r}_i} - \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \cdot \frac{\partial}{\partial \mathbf{p}_i} \right] \tag{B.2}$$

where $i = \sqrt{-1}$.

The equation (B.1) can be rewrite as $\mathrm{d}\boldsymbol{\gamma}/\mathrm{d}t = iL\,\boldsymbol{\gamma}$, which can be formally resolved as

$$\boldsymbol{\gamma}(t) = e^{iLt}\,\boldsymbol{\gamma}(0)\,, \tag{B.3}$$

where the operator $exp(iLt)$ is know as the *classical propagator* and defines a mapping in phase-space.

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